Literature Review

1 Introduction

This literature review examines existing research and technical solutions related to room booking, university timetable management, resource allocation, and real-time system updates. The purpose is to identify approaches that can be applied to the AITU Booking project and determine which requirements are not fully addressed by the reviewed sources.The review includes eight academic publications and four technical or practical sources. These sources cover room booking systems, timetable optimization, database functionality, real-time communication, PDF data extraction, and existing room scheduling services.

2 Analysis of Existing Research

Maulana et al. (2024) describe the development of the Paramadina Roomhub application for room booking. This research is relevant because it demonstrates how a digital system can support the room reservation process. However, the available information does not establish whether the system addresses all the requirements of AITU Booking, particularly real-time availability and concurrent booking requests.

Several studies focus on university course timetabling and optimization. Bashab et al. (2023) review optimization techniques, constraints, and open challenges in university timetabling. Müller et al. (2025) examine a real-world university timetabling problem, while Rappos et al. (2022) apply mixed-integer programming to timetable generation. Other studies explore hybrid teaching, individual study plans, and multi-criteria scheduling. These approaches are useful for understanding scheduling constraints, but AITU Booking is not intended to generate a new university timetable. Instead, it uses an approved timetable as input and manages room availability and consultation bookings.

Technical documentation provides solutions for specific implementation challenges. PostgreSQL range types can represent time intervals and help detect overlapping bookings. FastAPI WebSockets can deliver real-time updates to connected users, although additional database and application-level controls are required to prevent conflicting reservations. The pypdf library can extract text from PDF files, but imported timetable data must be validated because PDF layouts may be inconsistent. LibCal provides a practical example of a room and resource scheduling service.

Overall, the reviewed sources provide useful methods and technologies, but they do not establish that a single solution satisfies all the requirements of AITU Booking.

3 Comparison of Existing Solutions

Source	Approach	Main contribution	Limitations	Application to AITU Booking
Maulana et al. (2024)	Room booking application	Demonstrates a digital room reservation process	Does not establish coverage of all AITU-specific requirements	Reference for room booking functionality
Bashab et al. (2023)	Review of timetabling optimization	Summarizes constraints and optimization methods	Focuses on timetable optimization rather than room reservations	Helps identify scheduling constraints
Müller et al. (2025)	Real-world university timetabling	Examines practical timetable problems	Does not focus on consultation booking workflows	Provides context for university scheduling
Rappos et al. (2022)	Mixed-integer programming	Models and solves timetable problems	More suitable for timetable generation than booking management	Helps understand scheduling constraints
Davison et al. (2025)	Timetabling with hybrid teaching	Considers different teaching formats	Does not cover the complete AITU Booking workflow	Supports consideration of different room-use requirements
Steiner et al. (2025)	Curriculum-based timetabling	Considers individual study plans	Focuses on curriculum scheduling	Provides additional scheduling context
Dunke and Nickel (2023)	Multi-level, multi-criteria timetabling	Addresses complex scheduling objectives	Does not directly address real-time room booking	Helps understand multiple scheduling constraints
Chen et al. (2021)	Survey of university timetabling	Reviews trends and research opportunities	Broad overview rather than a specific booking	Provides background for the research

			solution	
PostgreSQL: Range Types	Database range operations	Supports the representation of time intervals	Does not independently implement the complete booking logic	Helps detect overlapping reservations
FastAPI: WebSockets	Real-time communication	Allows updates to be sent to connected clients	Does not guarantee booking consistency by itself	Supports real-time room availability updates
pypdf: Extract Text from a PDF	PDF text extraction	Extracts text from timetable documents	Complex layouts may require additional processing and validation	Supports timetable import
LibCal	Room and resource scheduling service	Provides an example of an existing scheduling solution	Its suitability for all AITU-specific workflows requires further evaluation	Serves as a practical reference for comparison

4 Research Gap

The reviewed sources address individual aspects of room booking and university scheduling, but they do not confirm that all AITU Booking requirements are covered by one solution. In particular, the project needs to combine importing an approved timetable from a PDF in the university's "10+2 weeks" format, displaying room availability, preventing conflicting reservations, scheduling staff consultations, and providing real-time updates.Another important requirement is the management of classroom keys through ID cards. The reviewed sources do not provide sufficient evidence that this workflow is covered by the examined approaches. Therefore, it should be treated as a project-specific requirement and validated with the university.The project must also ensure that simultaneous booking requests cannot create overlapping reservations. This requires database-level protection and application-level validation. The imported timetable must also be checked for correctness before it is used by the system.

5 Conclusion

The literature review shows that existing research and technologies provide a foundation for developing AITU Booking. Timetabling studies help identify scheduling constraints, while PostgreSQL, FastAPI WebSockets, and pypdf offer tools for managing time intervals, delivering updates, and importing timetable data.

However, the main challenge is to integrate these capabilities into one system that meets the university's specific requirements. AITU Booking will focus on using the approved timetable, managing room reservations and staff consultations, preventing booking conflicts, updating room availability in real time, and supporting the classroom key workflow through ID cards. These requirements define the main direction of the project.

References

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